

BILD 62: Last Day!



Class updates

- Code review due Friday at 5 pm
- SET evaluations by Saturday morning 🙏
- Projects due Sunday evening at 11:59 pm
 - Only one person needs to submit to *both* Canvas and the DataHub
- Final is on Monday at 7 pm (same room as lecture)
 - Final review session on Friday from 3-4:30 pm

Regardless of how you collaborate, in the end, everyone needs to submit via DataHub AND Canvas

so that we can
test your code

so that we can
leave comments

**No need for
data files;
ONLY ONE PER
GROUP!**

PLEASE INCLUDE YOUR DATA FILES!

Only one per group is fine

Final Exam Details

- The final exam for this class will be a similar format to the quizzes, with a mix of multiple choice, matching, and open ended questions.
- You will have 1.5 hours for this exam (it will not use the entire 3 hour time slot).
- One section of the final is related to *your* final project.

Study Sheet: To help you complete this final, you can bring one double-sided (standard size paper) sheet with you to the exam. This can be printed or handwritten. There are open questions on the exam that will ask you about code constructs (e.g., conditionals, functions, classes, loops, or modules) that you used in your final project, so **you are encouraged to leave yourself notes about components of your final project on your sheet.**

So, how did we do?



<https://www.menti.com/blwjz14ihb8g>

Course Objectives

- ★ Read and run basic Python programs, recognizing the structures used (i.e. variables, conditionals, loops, functions) and explaining how they work
- ★ Manipulate and create objects in Python, including data structures and classes
- ★ Write, edit, and execute Python code in Jupyter Notebooks as well as the command line
- ★ Visualize and ~~run hypothesis testing~~ on simple datasets in Python
- ★ Implement common algorithms for analyzing biological data (e.g., time series, images) and determine when such computations are appropriate

How can I keep
learning?



Competitions

Grow your data science skills by competing in our exciting competitions. Find help in the [documentation](#) or learn about [Community Competitions](#).

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[All competitions](#)
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[Research](#)
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[Analytics](#)
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New to Kaggle?



Try a Kaggle competition (or try working with data in your lab!)

Getting Started
14044 Teams

Getting Started
1274 Teams

2170 Teams

Topics and Subjects

Course Title	Course Description	Course Link	Next Course Dates
Computational Neuroscience	The curriculum integrates cutting-edge advances in machine learning and causality research with state-of-the-art modeling approaches in neuroscience. See prerequisites here.	See Course Details and Upcoming Dates	6-24 July, 2026
Deep Learning	Our Deep Learning (DL) course grew out of the realization that there is a real need for teaching an ethically responsible hands-on TA-guided code-first DL curriculum that emphasizes how DL can be used to advance science and achieve better scientific insights. See prerequisites here.	See Course Details and Upcoming Dates	6-24 July, 2026
Computational Tools for Climate Science	Through this program, students will gain skills and knowledge in the areas of climate science and impact, computational methods, data access, and scientific practices. See prerequisites here.	See Course Details and Upcoming Dates	13-24 July, 2026
NeuroAI	What are common principles of natural and artificial intelligence? The core challenge of intelligence is generalization. Neuroscience, cognitive science, and AI are all questing for principles that help generalization. Must take CN and DL or equivalent. See all prerequisites here.	See Course Details and Upcoming Dates	13-24 July, 2026
Open Science 101	Open Science 101 is a free, one-week, part-time course designed to introduce participants to the principles and practices of open science. This course is structured differently than our traditional full-time courses, view the course page for more information.	See Course Details and Upcoming Dates	No upcoming dates. Check back.

Neuromatch Courses

<https://neuromatch.io/courses/>

Learn Data Science

Whether you're new to the field or looking to take a step up in your career, Dataquest can teach you the data skills you'll need.

Learn Python, R, SQL, data visualization, data analysis, and

Take a FREE course!

[SIGN UP](#)

or

You can *keep learning online*

Python Basics for Data Analysis (Skill Path) or Data Scientist in Python (Career Path)

our [terms of use](#) and [privacy policy](#).

Other online courses

[Applied Computational Genomics @ U. of Utah - YouTube](#)

[MIT Deep Learning in Life Sciences - Spring 2021 - YouTube](#)

[Missing Semester IAP 2020 - YouTube](#)

<https://carpentries.org/workshop>

[Hypermodern Python · Claudio Jolowicz s/](#)

List curated by Fernando
Pozo ([Original Tweet](#))



bootcamp.extension.ucsd.edu/data/

UC San Diego, Division of Extended Studies Boot Camps > Data

DATA

Curriculum

GET PROGRAM INFO

First Name

NEXT

20%

Enrolling Now

CODING

- 08/22/22: Full-Time
- 09/19/22: Part-Time

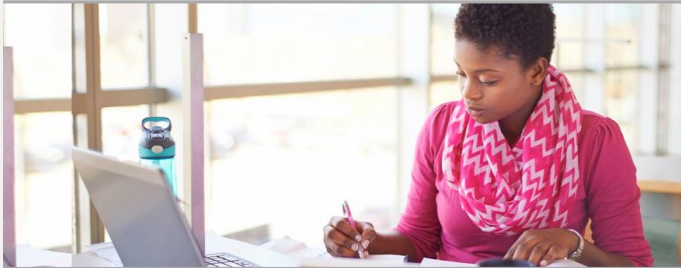
DATA SCIENCE

- 08/09/22: Part-Time

CYBERSECURITY

- 06/13/22: Part-Time

The Data Science and Visualization Boot Camp at UC San Diego, Division of Extended Studies



Explore: [Curriculum](#) | [Innovative Model](#) | [Comprehensive Support](#) | [How to Apply](#) | [Schedule](#)

Learn Data Science in 24 Weeks

Online San Diego Data Science and Visualization Boot Camp

The Data Science and Visualization Boot Camp at UC San Diego, Division of Extended Studies puts the student experience first, teaching students the knowledge and skills to conduct robust analytics.

dphi.tech/bootcamps

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Data Science Community Bootcamps

As a thriving global data science community, our bootcamps focus on learning and growing together. Join us to learn and enhance real-world data science skills from industry experts.

Learn For Free

Attended by Learners At

Google IBM HARVARD UNIVERSITY PayPal Yale KU LEUVEN

Upcoming Bootcamps Active Bootcamps

Sign up for a Data Science bootcamp

Create your own personal portfolio (on Github!)

Using Personal Projects to Land Your First Job in Tech

Nick Kolakowski

Jun 27, 2024

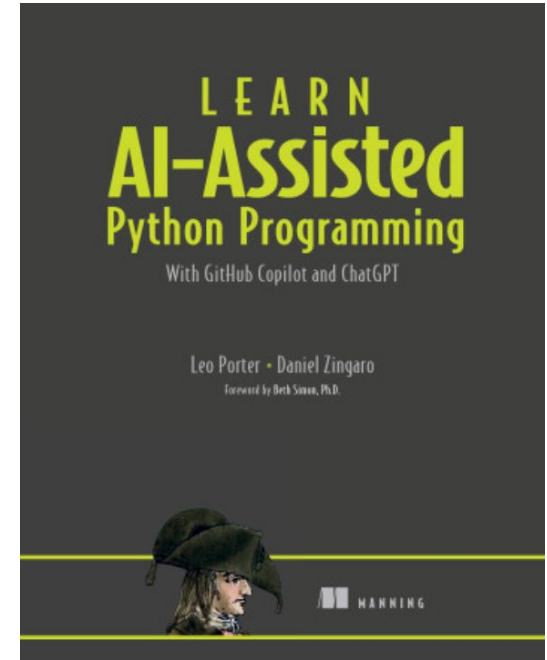
3 min read



From: <https://www.dice.com/career-advice/using-personal-projects-to-land-your-first-job-in-tech>

Continue learning alongside AI assistants

Just make sure you
understand what
the code is doing!!!!



Porter & Zingaro (2023)
[Learn AI-Assisted Python Programming](#) — entire
book on using Copilot &
VS Code!

Courses at UCSD

Biology classes

BIMM 143. Bioinformatics Laboratory (4). Bioinformatics is the analysis of big data in the biosciences. This course provides a hands-on introduction to the computer-based analysis of biomolecular and genomic data. Major topic areas include advances in sequencing technologies, genome resequencing and variation analysis, transcriptomics, structural bioinformatics, and personal genomics. This course will utilize free, web-based bioinformatics tools and no programming skills are required. Prerequisites: BILD 1 and BILD 4 or BIEB 123 or BIMM 101.

BIMM 149. Computation for Biologists (4). Course will provide students with the computational tools and problem-solving skills that are increasingly important to the biosciences. Students learn to program in a modern general-purpose programming language and write their own programs to explore a variety of applications in biology including simulations, sequence analysis, phylogenetics, among others. Prerequisites: BILD 1 and BILD 2.

BIPN 164 Computational Models and Theories in Neuroscience - New course - effective Winter 2022. This course covers mathematical models of neurons, synapses, and neural networks. We will introduce theoretical frameworks of brain activity and function to understand neural computation and control of behavior. We will discuss network dynamics, synaptic plasticity, learning and memory. Students will apply modeling approaches to address scientific questions and make predictions for experiments. Prerequisites: MATH 10A or MATH 20A and MATH 10B or MATH 20B and MATH 11

Biology classes (continued)

BIPN 162. Neural Data Science (4) Project-based course in which students will use computational notebooks to perform exploratory data analyses and to test hypotheses in large neuroscience data sets, including the differences between unique neuron types, leveraging text mining of the neuroscience literature, and human neuroimaging analyses. Prerequisites: MATH 11 and BIPN 140 and **BILD 62** or COGS 18 or CSE 8A

Neural Data Science (BIPN 162 or COGS 138)



By the end of this course, you'll be able to:

- Develop hypotheses specific to big data environments in neuroscience
- Design a big data experiment and excavate data from open sources
- Integrate data from multiple datasets to answer a biological question

Cognitive science classes

COGS 9. Introduction to Data Science (4). Concepts of data and its role in science will be introduced, as well as the ideas behind data-mining, text-mining, machine learning, and graph theory, and how scientists and companies are leveraging those methods to uncover new insights into human cognition.

COGS 118A. Supervised Machine Learning Algorithms (4). This course introduces the mathematical formulations and algorithmic implementations of the core supervised machine learning methods. Topics in 118A include regression, nearest neighborhood, decision tree, support vector machine, and ensemble classifiers. COGS 118A-B may be taken in either order. Prerequisites: COGS 18 or CSE 8B or CSE 11 and MATH 18 or MATH 31AH and MATH 20E and MATH 180A and COGS 108 or COGS 109 or COGS 118B or CSE 150 or CSE 151 or CSE 158 or ECE 174 or ECE 175A or consent of instructor.

COGS 118B. Introduction to Machine Learning II (4). This course, with COGS 118A, forms a rigorous introduction to machine learning. Topics in 118B include maximum likelihood estimation, Bayesian parameter estimation, clustering, principal component analysis, and some application areas. COGS 118A-B may be taken in either order. Prerequisites: CSE 8B or CSE 11 and MATH 18 or MATH 31AH and MATH 20E and MATH 180A or consent of instructor.

Cognitive science classes (*continued*)

COGS 118C. Neural Signal Processing (4). This course will cover theoretical foundations and practical applications of signal processing to neural data. Topics include EEG/field potential methods (filtering, Fourier (spectral) analysis, coherence) and spike train analysis (reverse correlation, spike sorting, multielectrode recordings). Some applications to neural imaging (optical microscopy, fMRI) data will also be discussed. Prerequisites: MATH 18 or MATH 31AH, COGS 14B or PSYC 60, and COGS 108 or COGS 109.

COGS 118D. Mathematical Statistics for Behavioral Data Analysis (4). Statistical methods for analyzing behavioral data. A mathematically sophisticated course covering both classical and Bayesian statistical methods for estimation, hypothesis testing, regression, and model comparison. Emphasis on both mathematical understanding of statistical methods as well as common applications. Prerequisites: MATH 18 or MATH 31AH and MATH 180A or consent of instructor.

COGS 119. Programming for Experimental Research (4). This course will help students in the behavioral sciences (cognitive science, psychology, linguistics, neuroscience, and related fields) learn how to program experiments and analyze and present data. Prerequisites: COGS 14B and MATH 18 or MATH 31AH and MAE 8.

COGS 138. Project-based course in which students will use computational notebooks to perform exploratory data analyses and to test hypotheses in large neuroscience datasets, including the differences between unique neuron types, leveraging text mining of the neuroscience literature, and human neuroimaging analyses. Prerequisites: COGS 18 and COGS 118C.

A path into advanced COGS classes...

Course name &
description

MATH 189

Quantitative
techniques for
analyzing big data

COGS 118A

Supervised
Machine Learning
Algorithms

COGS 118B

Introduction to
Machine
Learning

CSE 158

Data Mining

Prerequisites

MATH 18 or 20F or 31AH
MATH 20C
MATH 183

Uses R?

COGS 18
MATH 18
MATH 20E
MATH 180A
COGS 108/109

Uses Python

CSE 8B or CSE 11
MATH 18
MATH 20E
MATH 180A

DSC 40B
DSC 80
MATH 183

Visual Arts classes

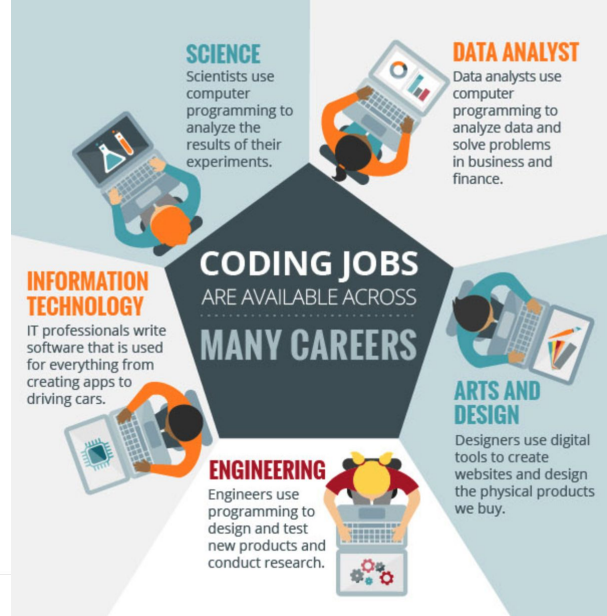
The **Interdisciplinary Computing and the Arts major** draws upon and aims to bring together ideas and paradigms from computer science, art, and cultural theory. <https://visarts.ucsd.edu/undergrad/major-req/icam.html>

https://visarts.ucsd.edu/_files/curriculums/ICAM%20Curriculums/ICAM-Curriculum-Sheet-FA21-and-Later1.pdf

Careers after BILD 62

Why should I learn how to code?

- Coding is useful for:
 - Data acquisition (controlling hardware, image acquisition, etc)
 - Data analysis & visualization
 - Computational modeling
- Beyond research, there are more and more jobs for software engineers, especially those with specialized knowledge →



nature

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CAREER Q&A | 31 May 2022

Why science needs more research software engineers

Ten years after their profession got its name, research software engineers seek to swell their ranks.

By [Chris Woolston](#)

<https://www.nature.com/articles/d41586-022-01516-2>

[Home](#) ▸ [Services](#) ▸ Research Software Engineering

Princeton University Research Software Engineering Summer Fellows Program - 2026

Call for Applications

Are you an undergraduate or graduate student enrolled at a [US-accredited College or University](#) [?](#) Are you enthusiastic about software & programming and interested in applying your skills to exciting academic research problems? Then the Princeton University Research Software Engineering Summer Fellows program could be for you. This program, supported by the [Alfred P. Sloan Foundation](#) [?](#), enables summer Fellows to work under the mentorship of a professional Research Software Engineer (RSE) to build, develop, and optimize software used in cutting-edge Princeton research software projects.

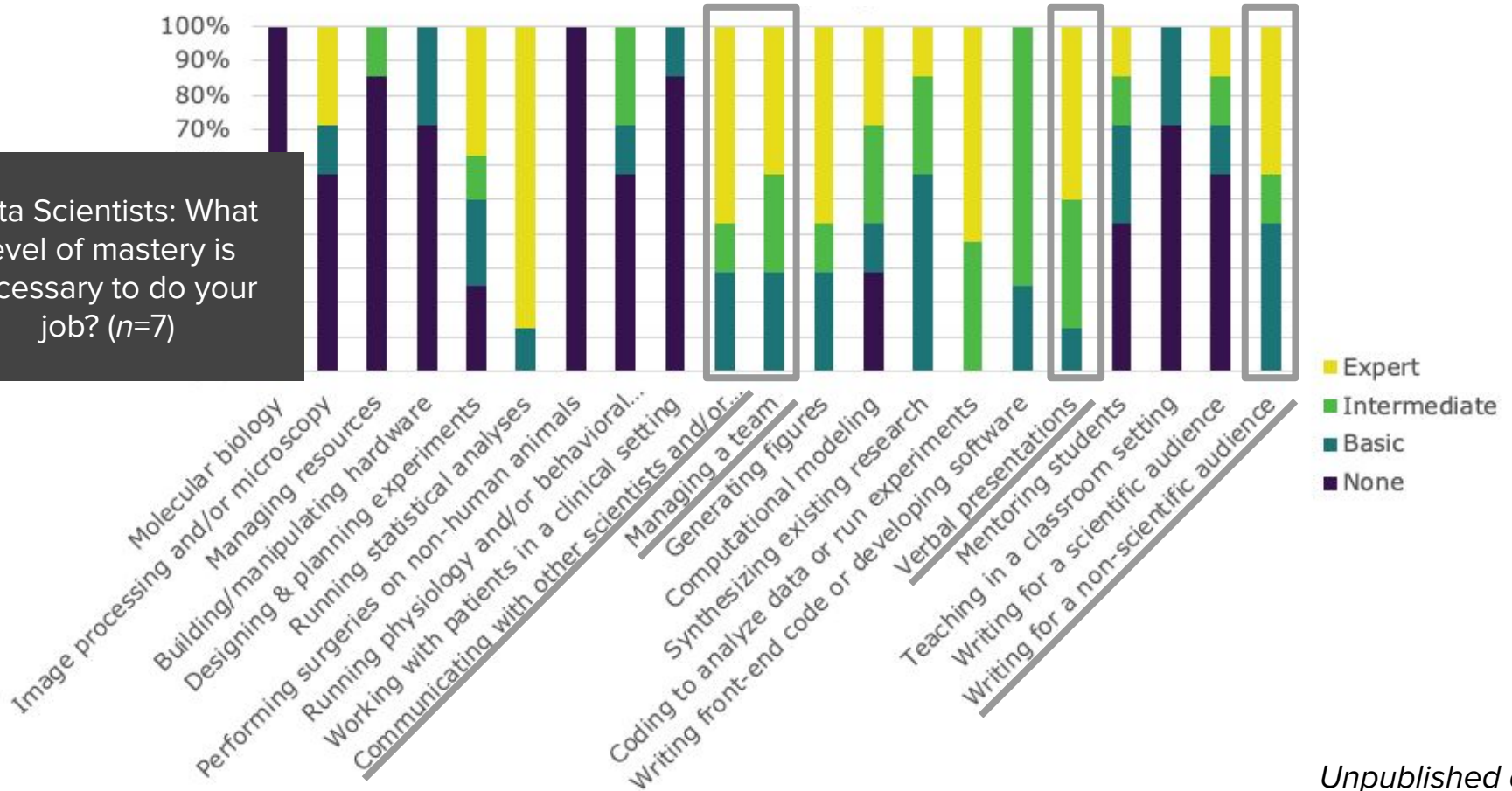
The RSE Fellows Program lasts 10 weeks during the summer starting as early as June 1 and ending some time in August. There is some flexibility in starting and ending dates to accommodate Fellows from institutions with varying academic year schedules. Each Fellow is expected to be available full-time, not employed in any other activity (including graduate assistantships) or following significant coursework. **The fellowship program is primarily a remote experience** - Fellows will remain in their home location and work remotely with their mentor. Summer Fellows will receive a training stipend of \$720/week, paid monthly, during the program, assuming satisfactory progress. The program provides additional funds for a one-week site visit at Princeton University for all Fellows. The site visit is optional, but highly encouraged. During this time, Fellows have the opportunity to interact with their mentor and with other RSEs, students, and potentially other Fellows. Funding is available for Fellows who successfully complete the program to optionally attend [USRSE'26](#) [?](#) along with their mentors (technical submissions by Fellows to the conference are encouraged but not required).

[Research Software Engineering](#)[RSE Partnership Guide](#)[Group Members](#)[RSE projects](#)[RSE Blog](#) [?](#)[Princeton Regional Group](#)[2026 Research Software Engineering](#)

<https://researchcomputing.princeton.edu/services/research-software-engineering/2026-research-software-engineering-summer-fellows-program>

Other skills are still important!!!

Data Scientists: What level of mastery is necessary to do your job? (n=7)

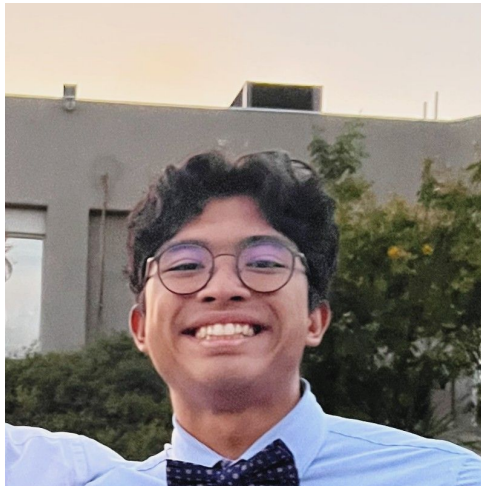


Unpublished data

Thanks!



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Thanks for a
great quarter.

```
learn how  
to code  
while  
looking at  
> genes  
> cells  
> brains
```